

DamageBDD Mobile Platform Multiplayer Behaviour Verification and On-Chain Social Proof System

DamageBDD

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Abstract

DamageBDD is building the world's first mobile-first Behaviour Driven Development (BDD) platform with real-time streaming execution, multiplayer collaboration, and an on-chain social verification layer powered by Aeternity smart contracts and the Damage Token (AEX-9). This document outlines the product vision, on-chain architecture, market opportunity, revenue model, and a concrete development roadmap for investor review.

Contents

1 Executive Summary

DamageBDD introduces a new category of software engineering infrastructure: a **real-time, multiplayer, cryptographically verified behaviour platform** that blends:

- Mobile-first BDD creation
- AI-assisted step completion
- Real-time streaming execution visualisation
- Multiplayer collaboration and competitive verification
- Fully on-chain reputation, identity, and economic incentives

We propose a disruptive transformation of the global QA, audit, verification, and compliance industries. The platform turns software correctness into a social, verifiable, economically-incentivised experience that can run on any mobile phone, anywhere in the world.

DamageBDD is simultaneously:

1. A developer tool,
2. A verification marketplace,
3. A crypto-economic coordination engine,
4. A thin-mobile client for deterministic proofs of behaviour.

At scale, DamageBDD becomes the **verification layer for the internet**.

2 Product Overview

2.1 Vision

Make it possible for any human, anywhere, on any mobile phone, to:

1. Describe how software *should* behave in human-readable English.
2. Execute that behaviour with a single tap.
3. Stream real-time, animated verification results.
4. Earn rewards for successful verification.
5. Build provable reputation on-chain.
6. Join a multiplayer arena of software testing.

Software reliability becomes a social experience, not a corporate silo.

2.2 Core Differentiators

- **Mobile-first BDD editor** — thumb-optimised, keyword-aware, and AI-assisted.
- **Streaming execution engine** — real-time progress visualised as cinematic animations.
- **Multiplayer verification** — multiple users running the same feature synchronously.
- **On-chain social layer** — claps, comments, forks, runs, identity, performance.
- **Economic incentives** — DAMAGE token for payouts, stakes, bounties, and proofs.
- **Decentralised proof-of-behaviour** — every result secured by blockchain.

3 Mobile BDD Editor

3.1 Interface Principles

The editor emphasises:

- Maximum ergonomic usage with a single thumb.
- Zero learning curve for non-technical contributors.
- Real-time suggestions from step libraries.
- Smooth fallback to raw Gherkin mode for experts.

3.2 Features

1. **Keyword completion:** Given / When / Then / And chips.
2. **Step templates:** Auto-insert HTTP steps, variable steps, JSON assertions.
3. **Smart autocompletion:** Matching pre-existing variables.
4. **Scenario builders:** One-tap generation of common scenario structures.
5. **Syntax highlighting:** Mobile-friendly monospaced layout with gutters.

3.3 Execution

Upon running the scenario:

- The client posts the feature to a DamageBDD endpoint.
- Real-time streaming begins (via Cowboy HTTP chunking).
- Each step animates on the screen (pending → active → pass/fail).

4 Multiplayer Verification

4.1 Sessions

When multiple users execute the same `feature_hash`, the execution becomes a **shared session**. The UI displays:

- Avatars of participants
- Per-step grids showing how each player progressed
- Mixed pass/fail indicators
- Live updates

4.2 Arena Mode

The Arena is a global and local social environment:

- Featured tests trending by run count
- Active challenges with bounties
- Personal squads formed through co-verification
- Fork trees showing descendant scenarios

4.3 Competitive Mechanics

Verification becomes a game-like environment:

- On-chain “first verifier” bonuses
- Time-to-verification leaderboards
- Co-op debugging sessions
- Bounties for bug discovery

5 On-Chain Architecture

5.1 Contracts

5.1.1 Profile Contract

Maps:

$$\text{account} \rightarrow \{handle, avatar_cid, socials, reputation_scores\}$$

Provides decentralised identity linking.

5.1.2 FeatureRuns Contract

Each run publishes:

```
{feature_hash, account, status, damage_spent, payout_sats, timestamp}
```

This is the canonical source of truth for all verification results.

5.1.3 Reactions Contract

Supports:

```
clap, comment(ipfs_cid), fork, share(platform)
```

5.2 Immutable Proofs

Everything is reconstructable from on-chain events:

- Feature popularity
- Developer productivity
- Historical correctness
- Reputation and trust scores
- Challenge participation

5.3 Economic Layer

Powered by the **DAMAGE Token (AEX-9)**:

- Verification fees
- Bounties
- Staking incentives
- Anti-spam economics
- Revenue share for the network

6 Social Media Integration

6.1 On-Chain Tracking of Off-Chain Shares

While posting to X/LinkedIn is done via native share sheets, the *fact* of sharing is stored on-chain for reputation:

```
share(feature_hash, account, platform)
```

6.2 Comments & Discussions

All comments store text on IPFS and reference it via:

```
comment(feature_hash, account, ipfs_cid)
```

6.3 Fork Trees

Forking a feature creates a lineage:

```
fork(parent_hash, child_hash, account)
```

This creates a long-term evolutionary tree of BDD specifications for a domain.

7 Market Opportunity

7.1 Industry Pain Points

- Software failure costs companies trillions annually.
- QA is underfunded and often treated as a cost centre.
- Verification lacks transparency and accountability.
- Remote contributors have no reliable reputation currency.
- Trust in software claims is low.

7.2 DamageBDD Advantages

- Globally crowdsourced verification.
- Pay-per-proof model.
- Immutable audit trail of correctness.
- Strong incentives for quality outcomes.
- A mobile-first distribution strategy for global reach.

7.3 Target Customers

- Enterprise QA teams
- Blockchain projects seeking real verification
- Fintech, healthcare, compliance-heavy sectors
- Open-source maintainers
- Freelance testers and global verification labour

8 Business Model

8.1 Revenue Streams

1. **Verification Fees** Companies pay in DAMAGE or BTC for verifications.
2. **Marketplace Bounties** Users complete challenges and receive rewards.
3. **Reputation Economy** High-ranked verifiers receive premium payout rates.
4. **Enterprise Licenses** Private networks, custom integrations, compliance suites.
5. **Token Appreciation** As utility grows, token demand increases naturally.

9 Roadmap

9.1 Phase 1 — Mobile BDD MVP (0–3 Months)

- Mobile-first BDD editor
- Step completion library
- Feature execution with streaming results
- Basic wallet integration
- Minimal identity system

9.2 Phase 2 — On-Chain Verification Layer (3–6 Months)

- Deploy Profile, FeatureRuns, Reactions contracts
- On-chain result publishing
- Visual live run sessions
- Tokenised payouts

9.3 Phase 3 — Multiplayer Arena (6–9 Months)

- Challenges and bounties
- Coop/competitive verification modes
- Fork trees and collaborative debugging
- Reputation ranking algorithms

9.4 Phase 4 — Enterprise Expansion (9–18 Months)

- Compliance modules
- Regulatory-grade reporting
- High-concurrency load verification
- Integration marketplace

10 Investment Request

DamageBDD is seeking strategic investment to accelerate development and scale the platform globally.

10.1 Use of Funds

- Core engineering team expansion
- Mobile development (iOS and Android)
- Smart contract engineering and audits
- Infrastructure scaling
- Marketing, community, and enterprise outreach

10.2 Investor Benefits

- Early token allocation
- Equity stake or revenue share options
- Governance participation
- Access to enterprise verification data analytics

11 Conclusion

DamageBDD is not merely a developer tool. It is:

- A verified computing platform,
- A crypto-economic coordination engine,
- A global multiplayer QA network,
- A new digital labour market,
- And the foundation of a future *Proof of Behaviour* standard.

This proposal outlines a path to build the verification layer for the modern software world. With strategic investment, we can expand from a powerful working product into a global standard for correctness, accountability, and trust.

DamageBDD

<https://damagebdd.com>